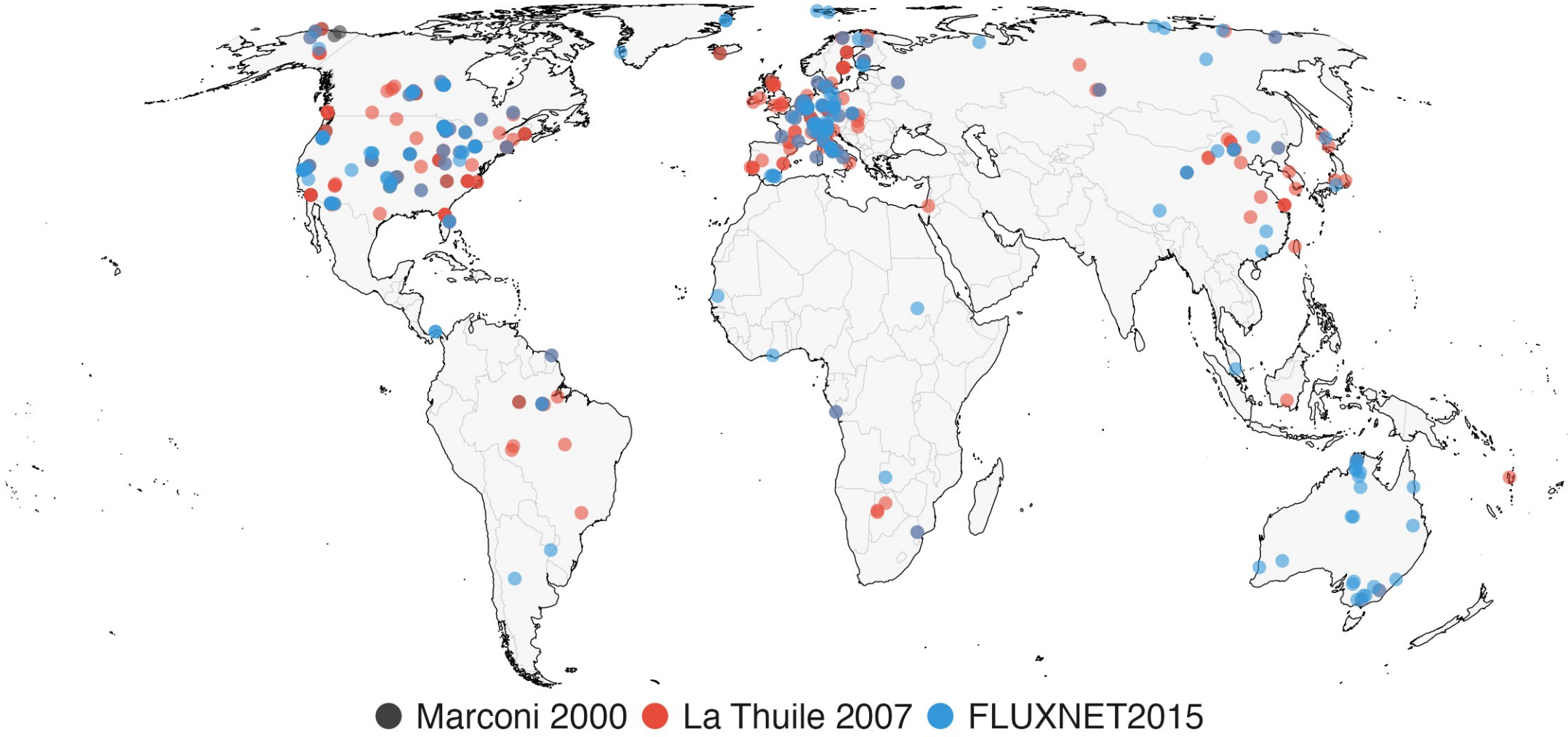
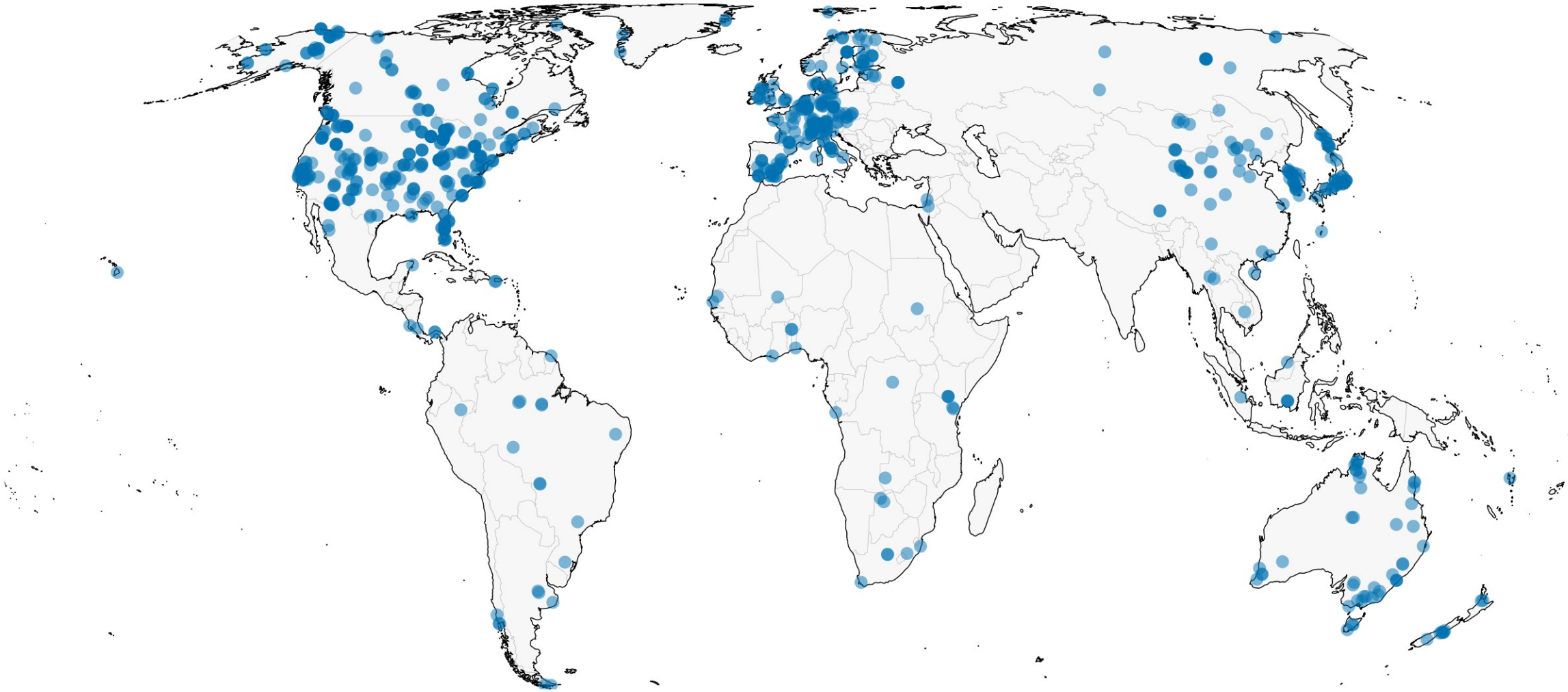
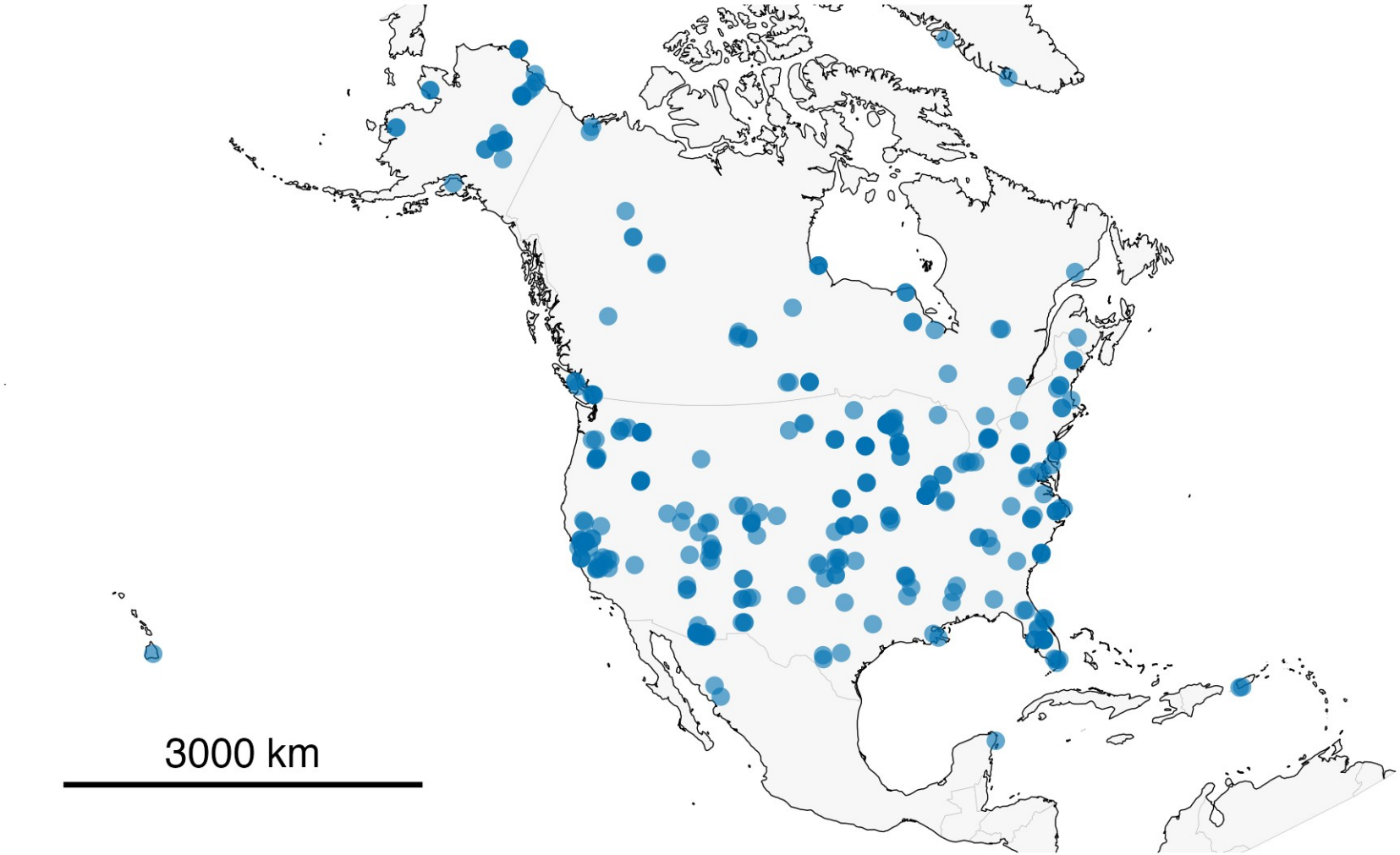
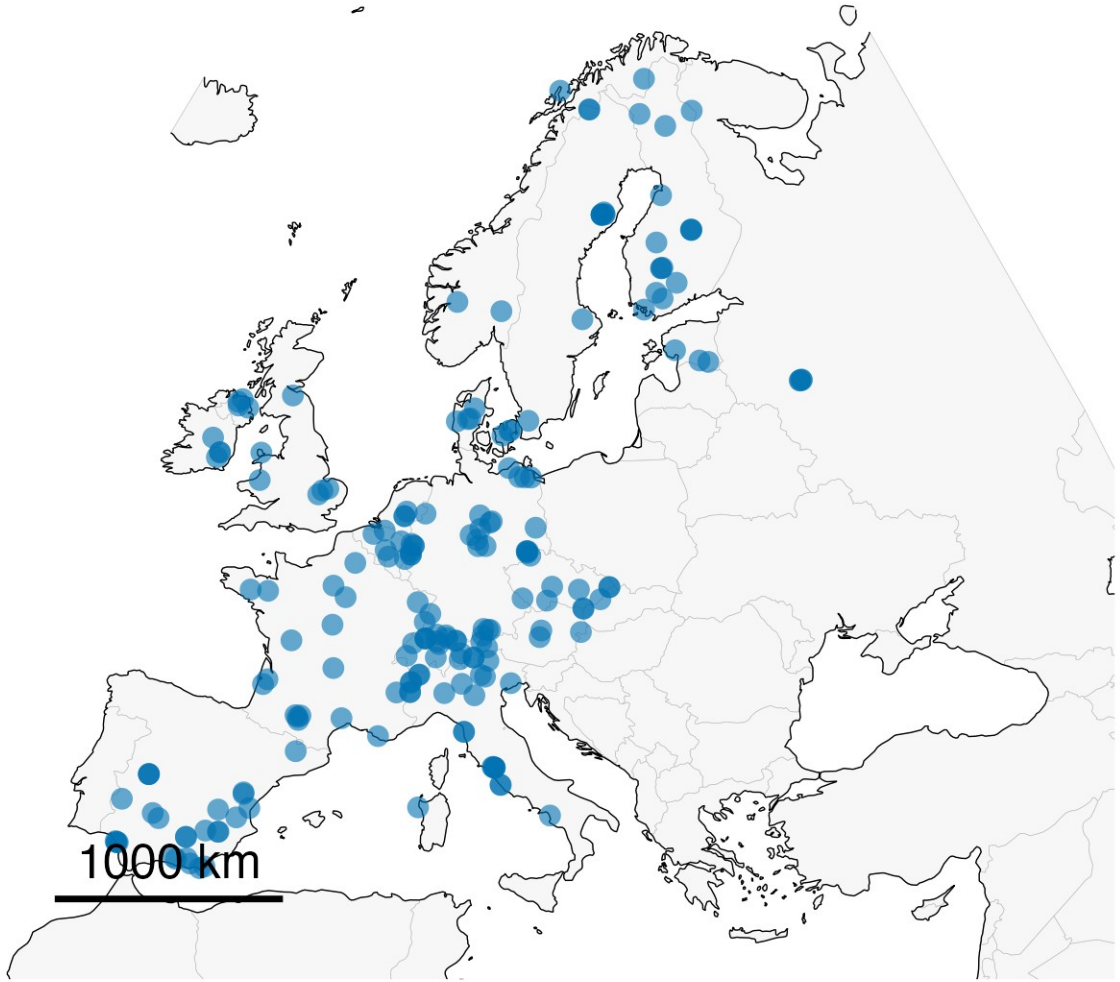
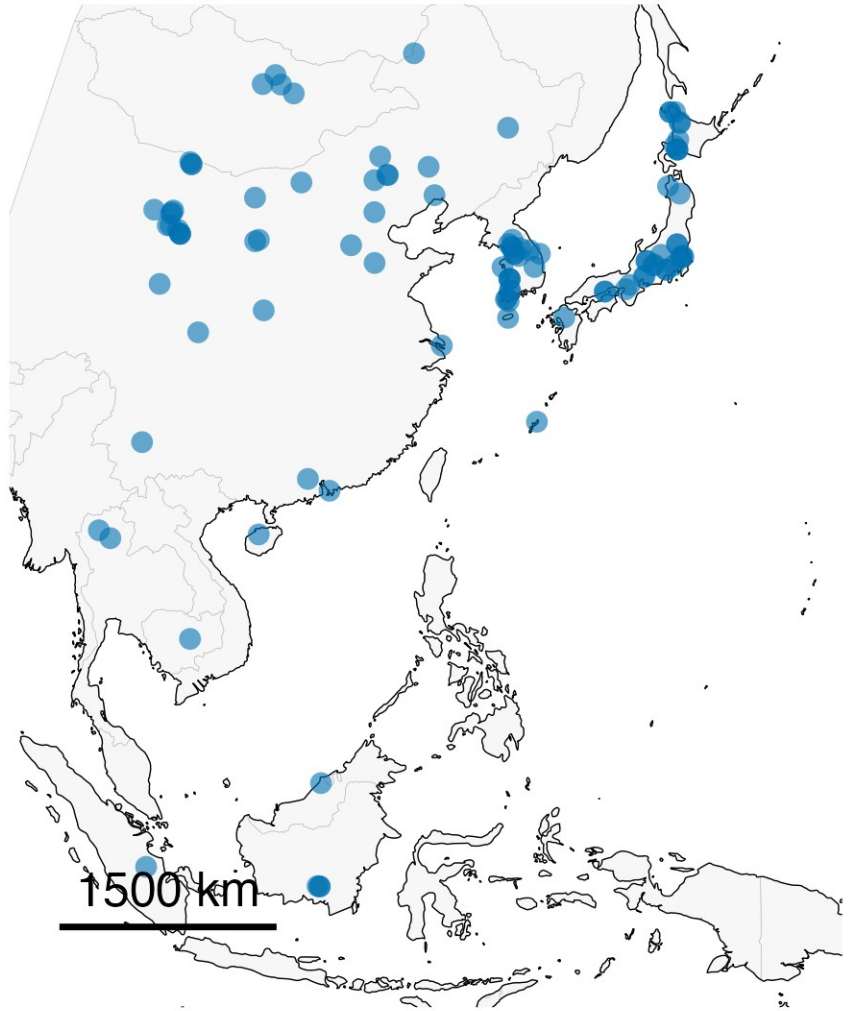
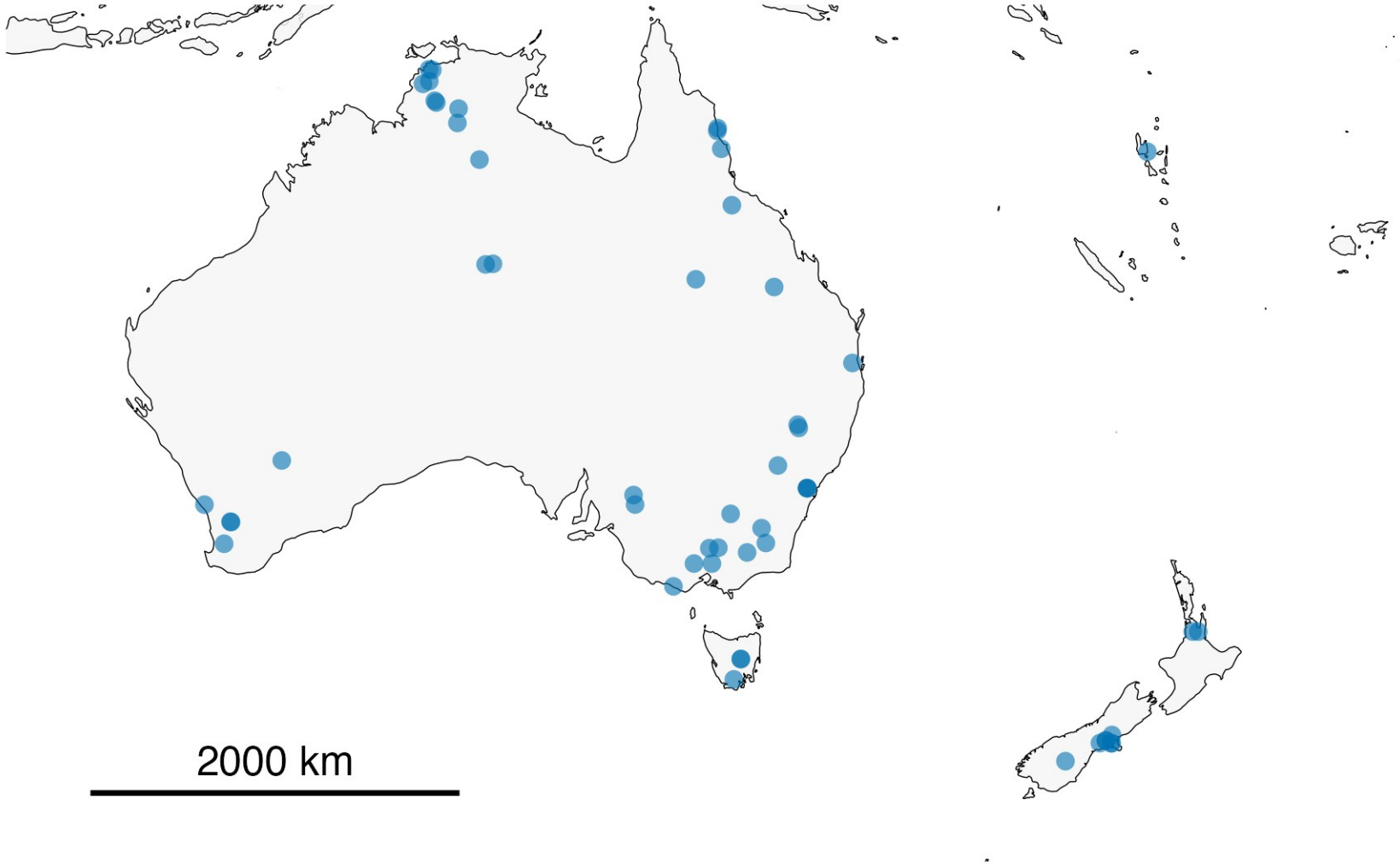

Marconi 2000
La Thuile 2007
FLUXNET2015

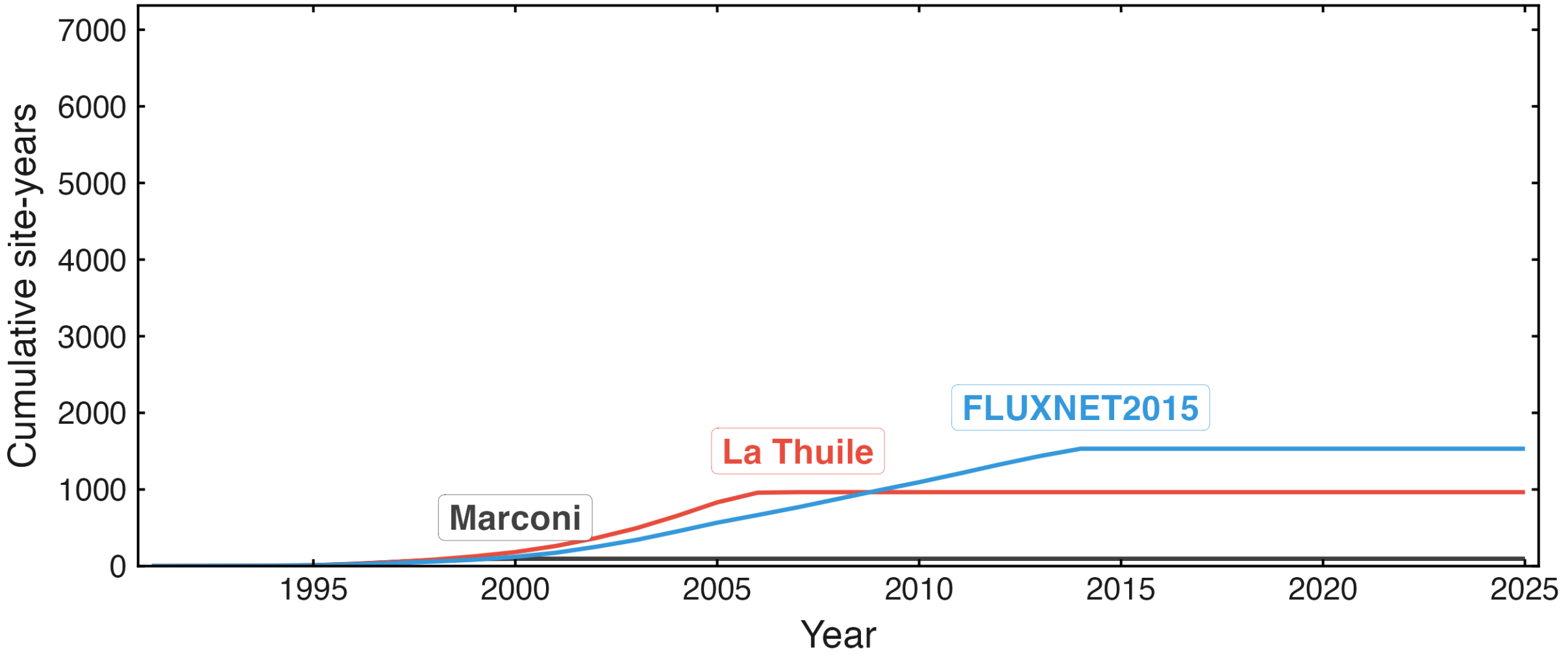


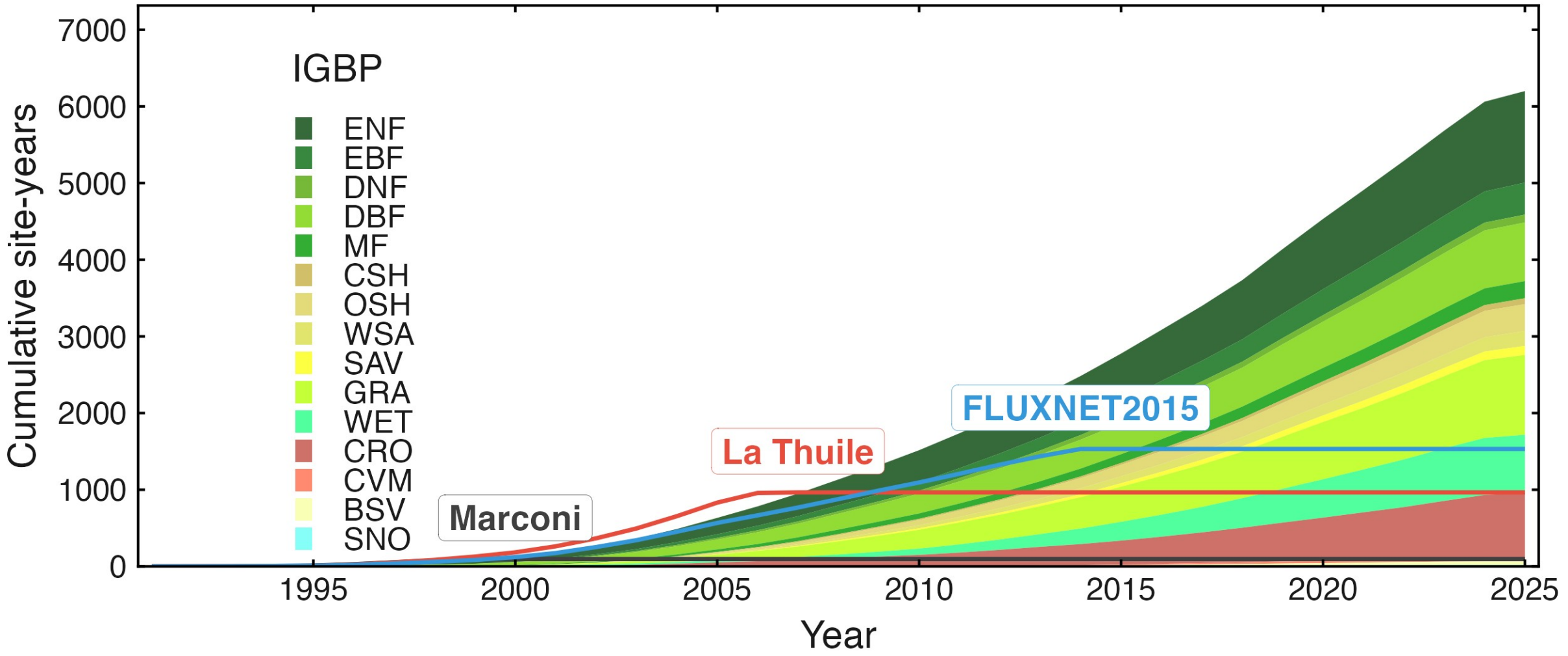

3000 km

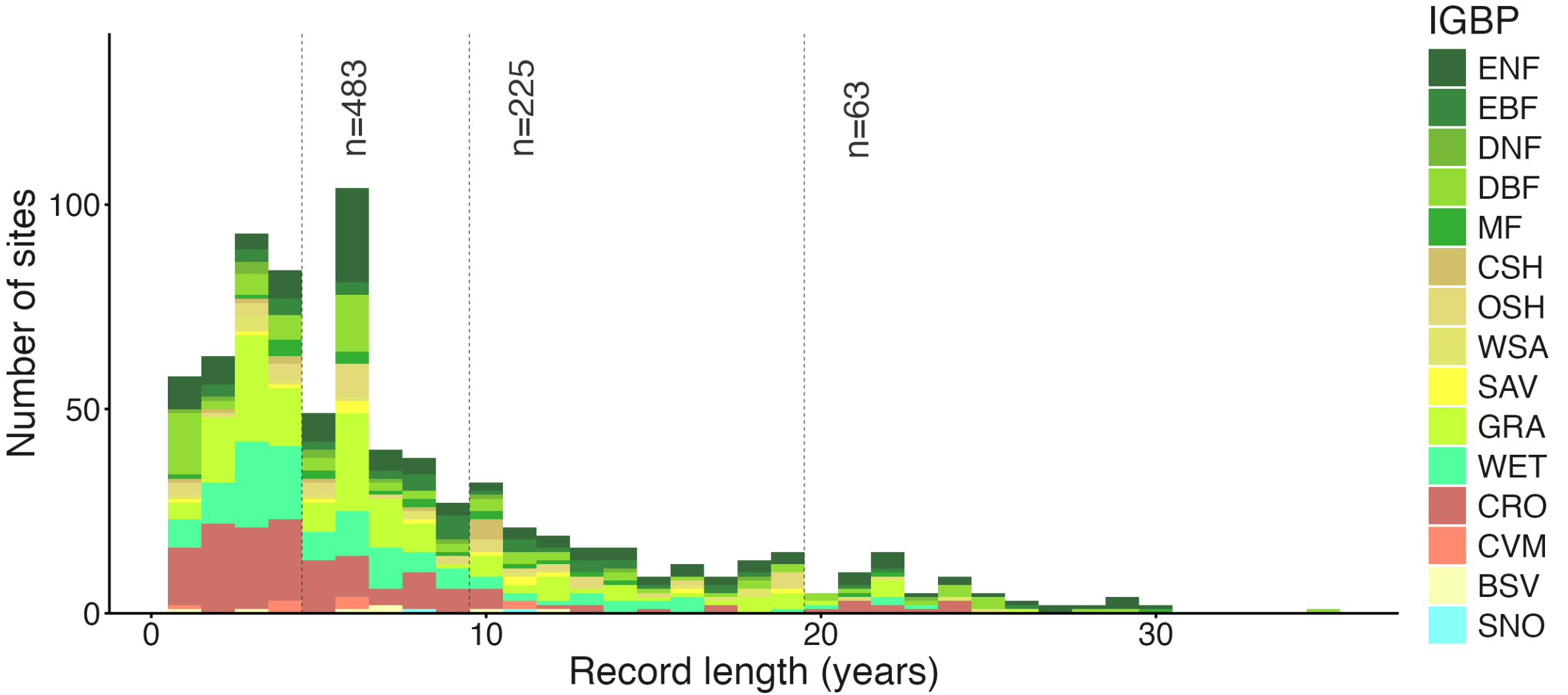

1000 km

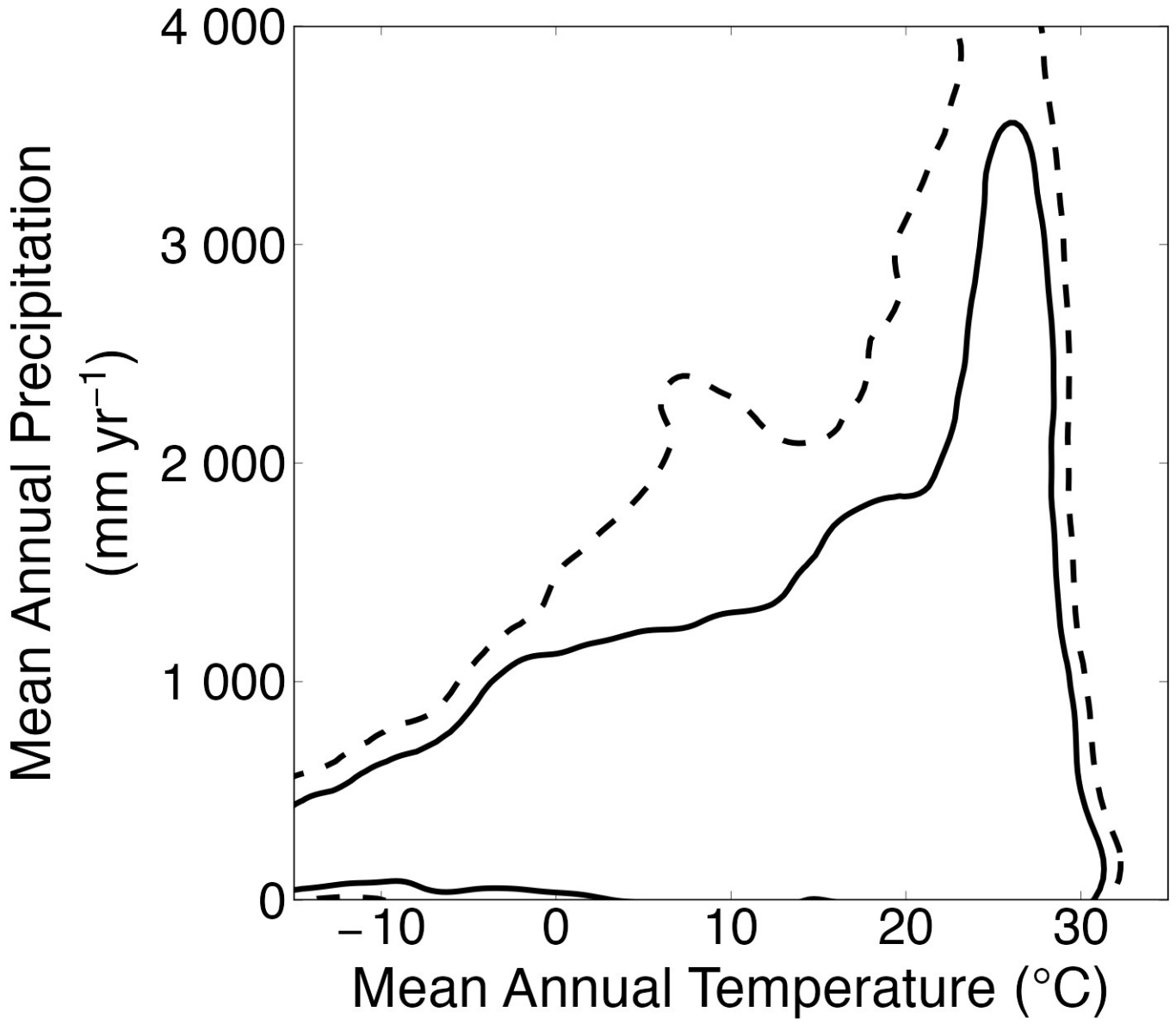

1500 km

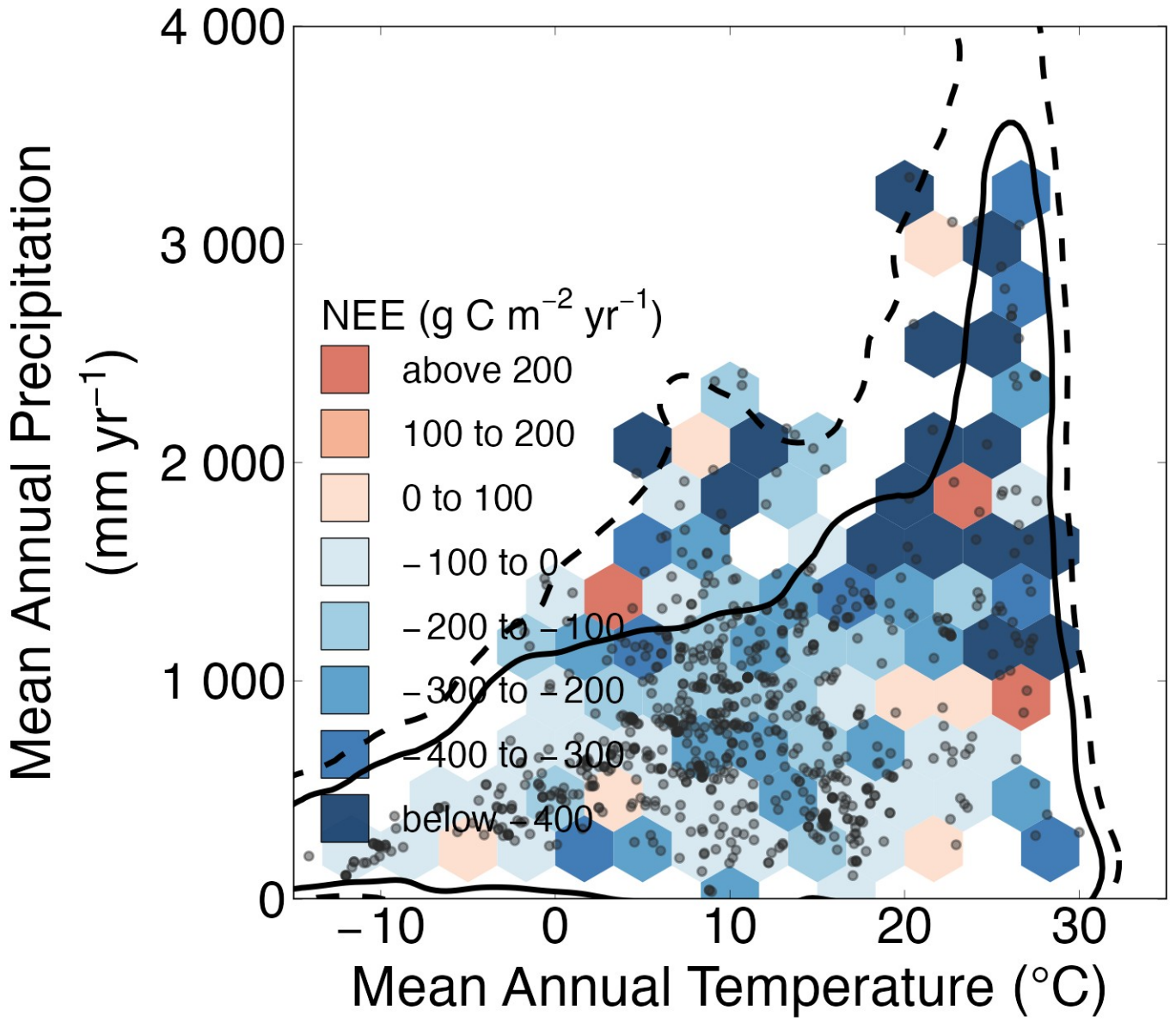

2000 km

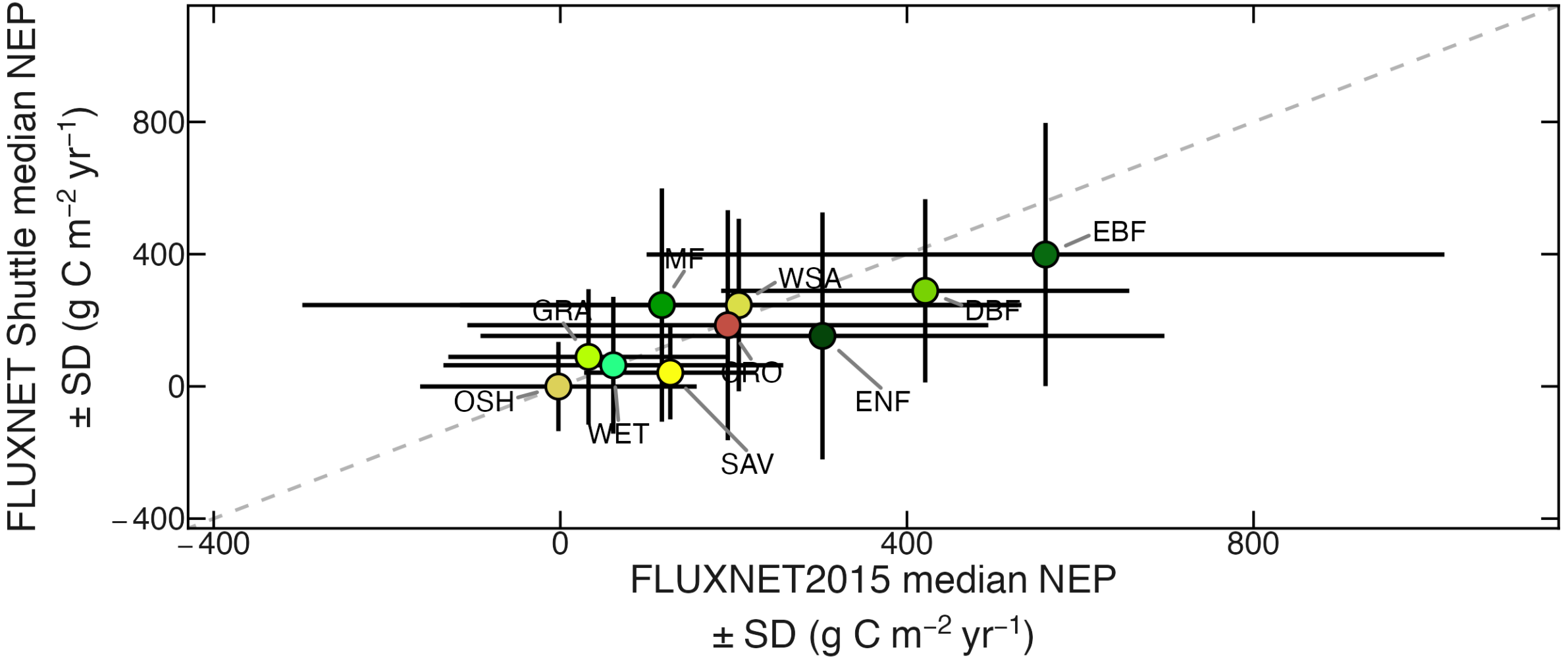

Cumulative site-years
7000
6000
5000
4000
3000
2000
1000
0
1995
2000
2005
2010
2015
2020
2025
Year
Marconi
La Thuile
FLUXNET2015

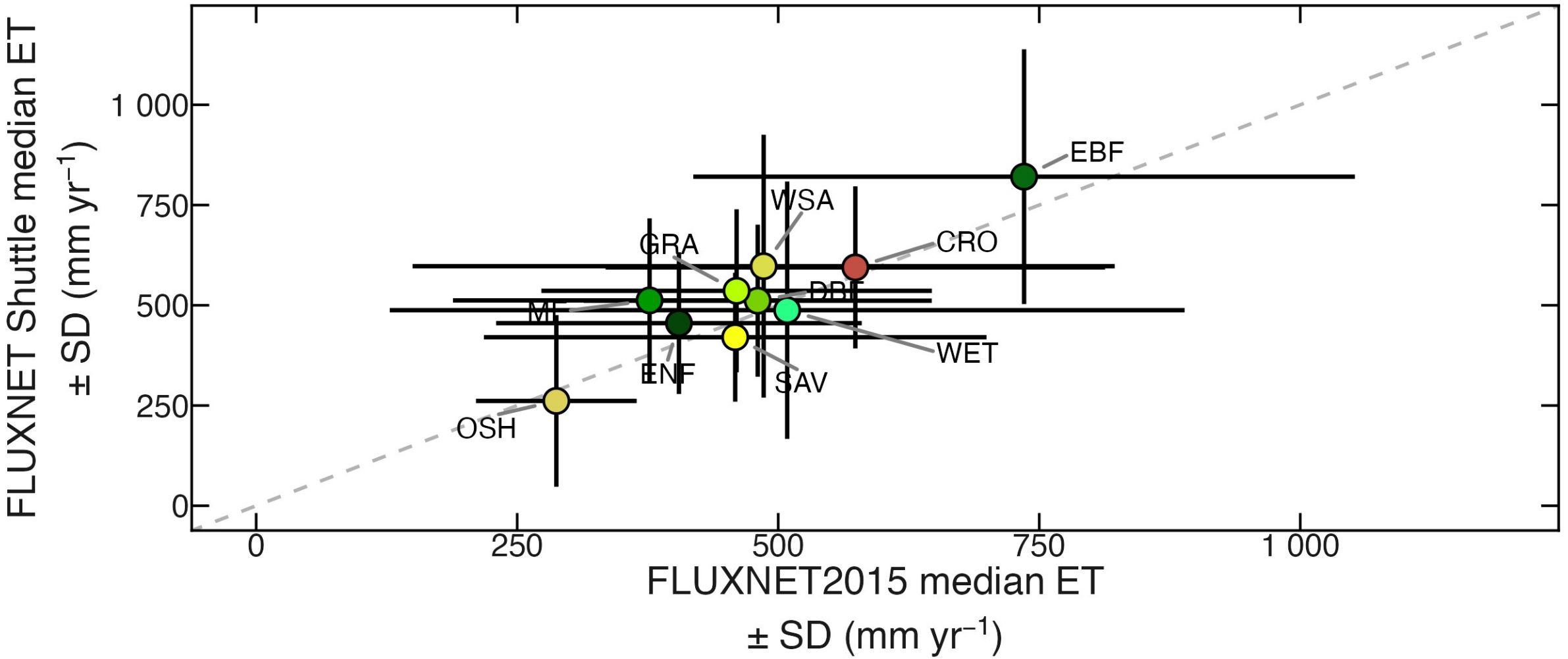

IGBP
ENF
EBF
DNF
DBF
MF
CSH
OSH
WSA
SAV
GRA
WET
CRO
CVM
BSV
SNO
Marconi
La Thuile
FLUXNET2015
Cumulative site-years
Year
0
1000
2000
3000
4000
5000
6000
7000
1995
2000
2005
2010
2015
2020
2025

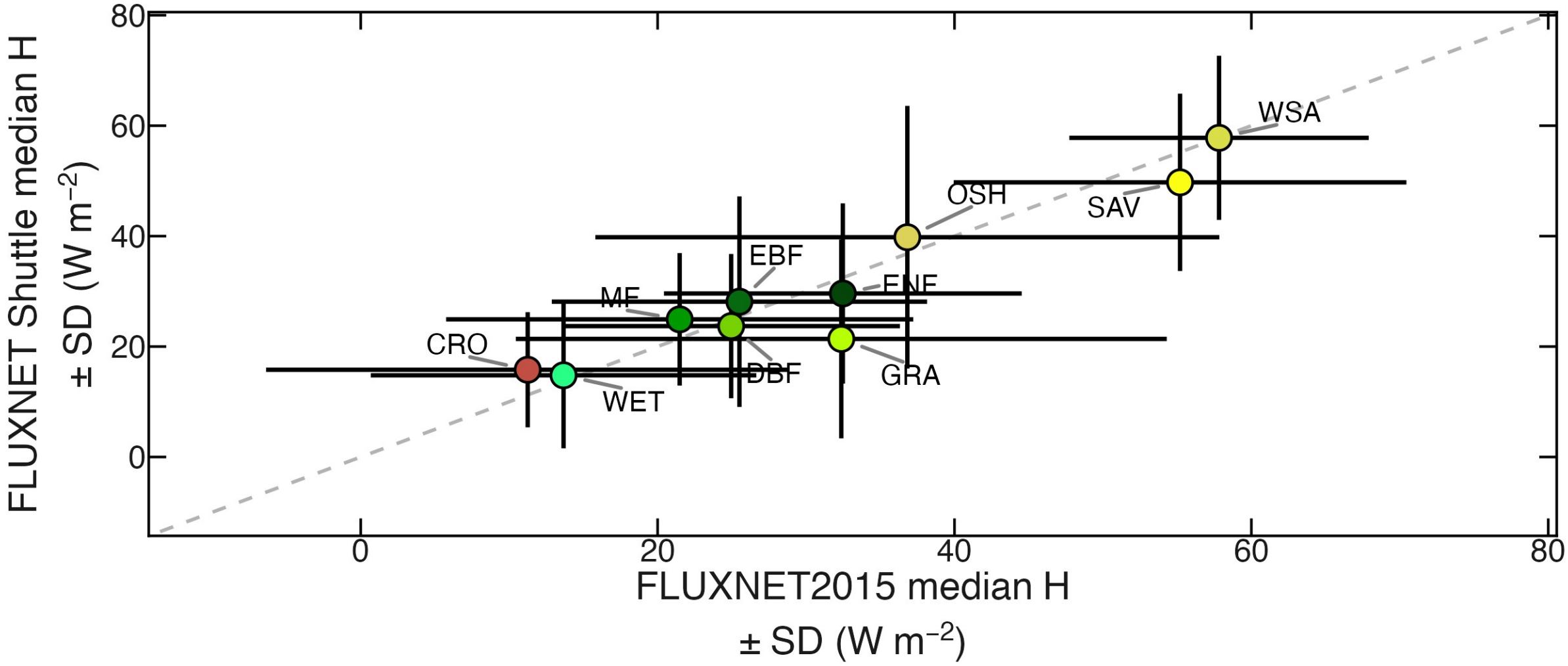

n=483
n=225
n=63
Number of sites
100
50
0
0
10
20
30
Record length (years)
IGBP
ENF
EBF
DNF
DBF
MF
CSH
OSH
WSA
SAV
GRA
WET
CRO
CVM
BSV
SNO

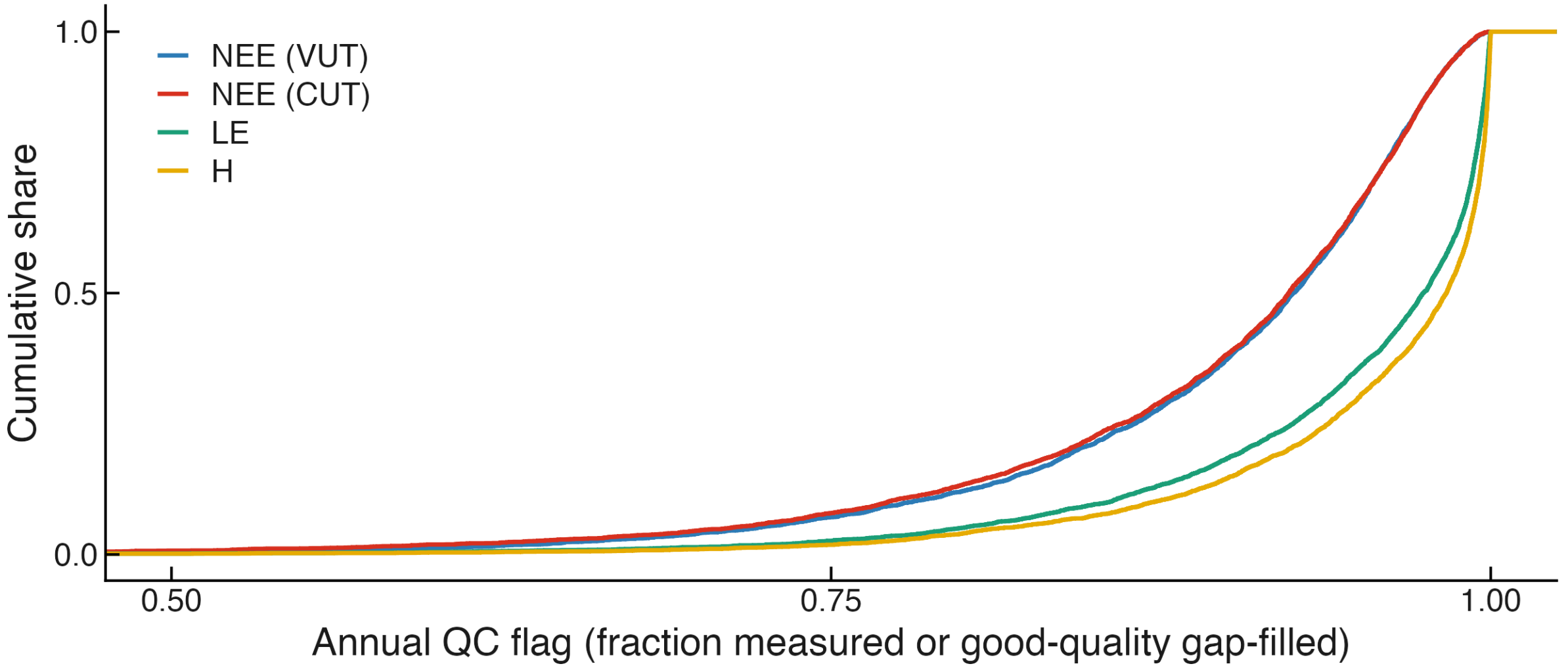

4 000
3 000
2 000
1 000
0
Mean Annual Precipitation
(mm yr^{-1})
−10
0
10
20
30
Mean Annual Temperature (°C)

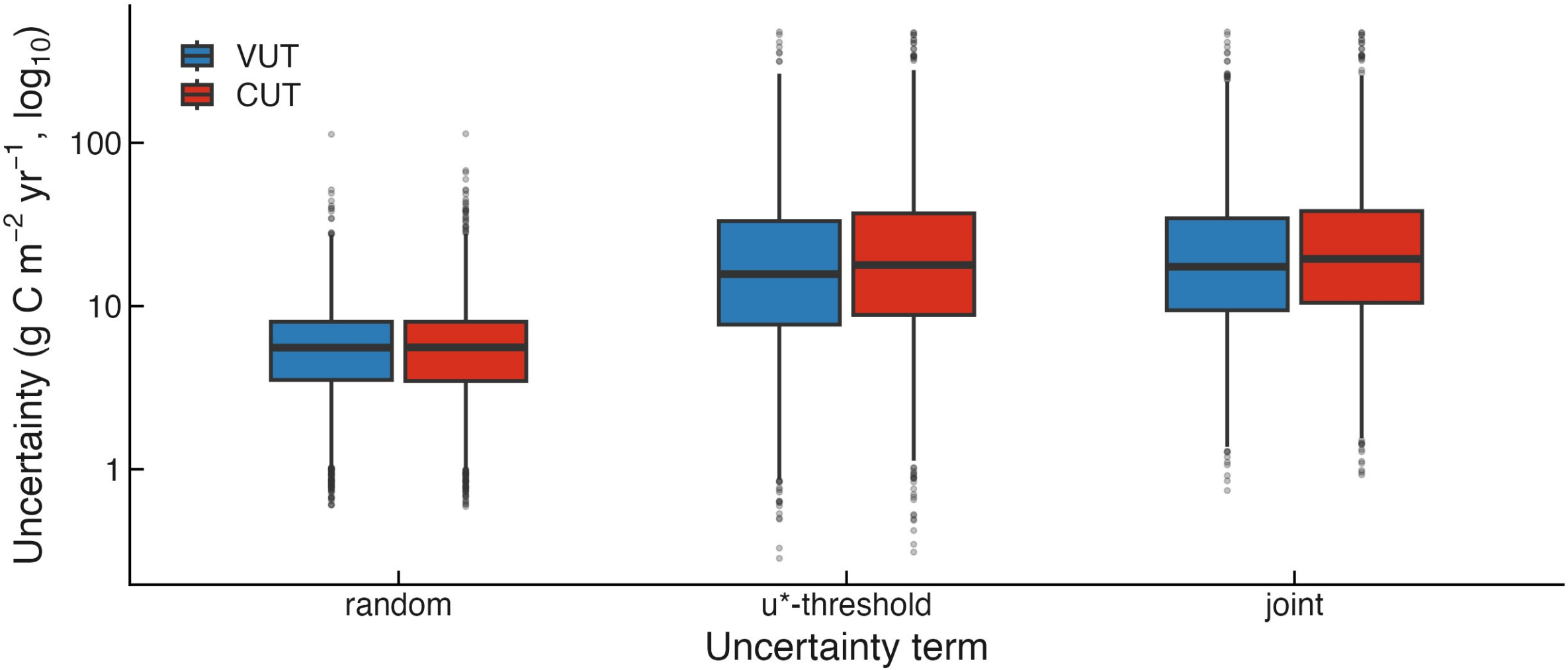

Mean Annual Precipitation
(mm yr^{-1})
4 000
3 000
2 000
1 000
0
NEE (g C m^{-2} yr^{-1})
above 200
100 to 200
0 to 100
−100 to 0
−200 to −100
−300 to −200
−400 to −300
below −400
−10
0
10
20
30
Mean Annual Temperature (°C)

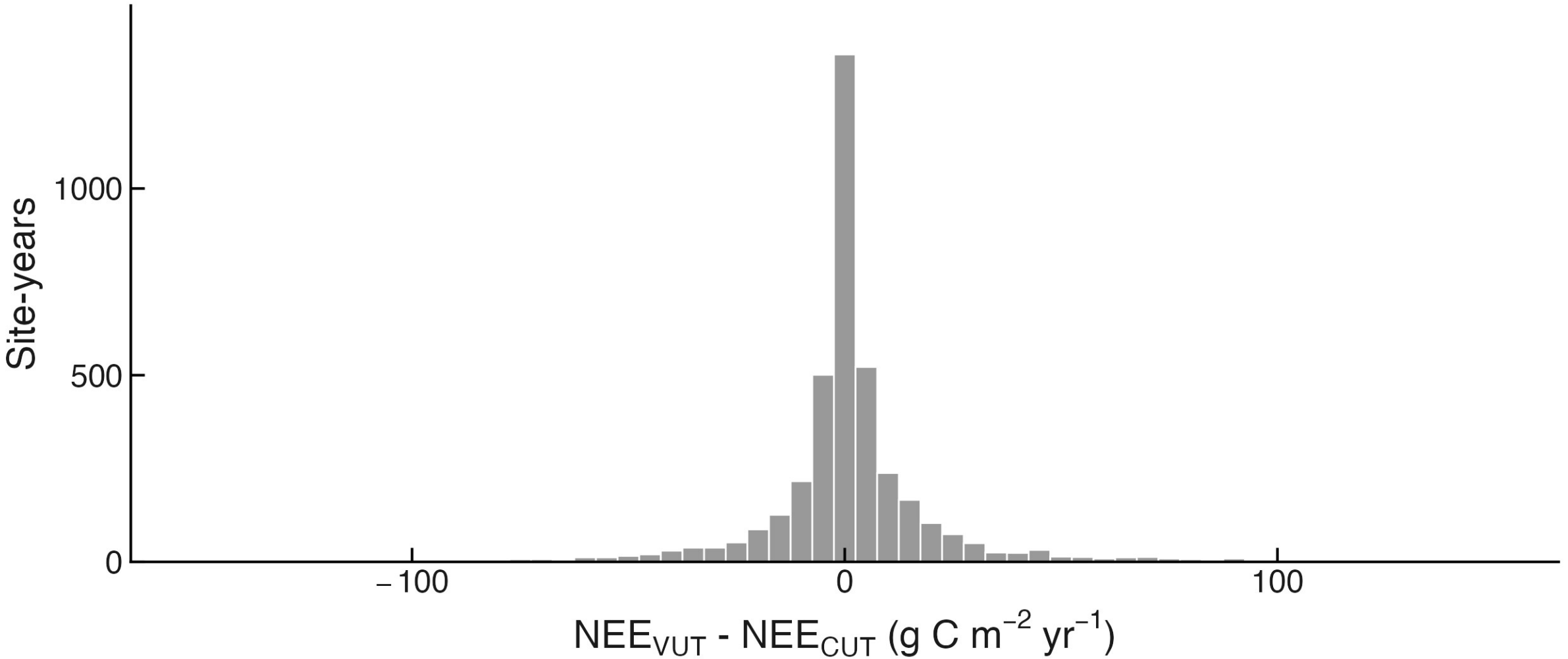

FLUXNET Shuttle median NEP
± SD (g C m−2 yr−1)
800
400
0
−400
−400
0
400
800
FLUXNET2015 median NEP
± SD (g C m−2 yr−1)
EBF
MF
WSA
GRA
DBF
CRO
OSH
ENF
WET
SAV

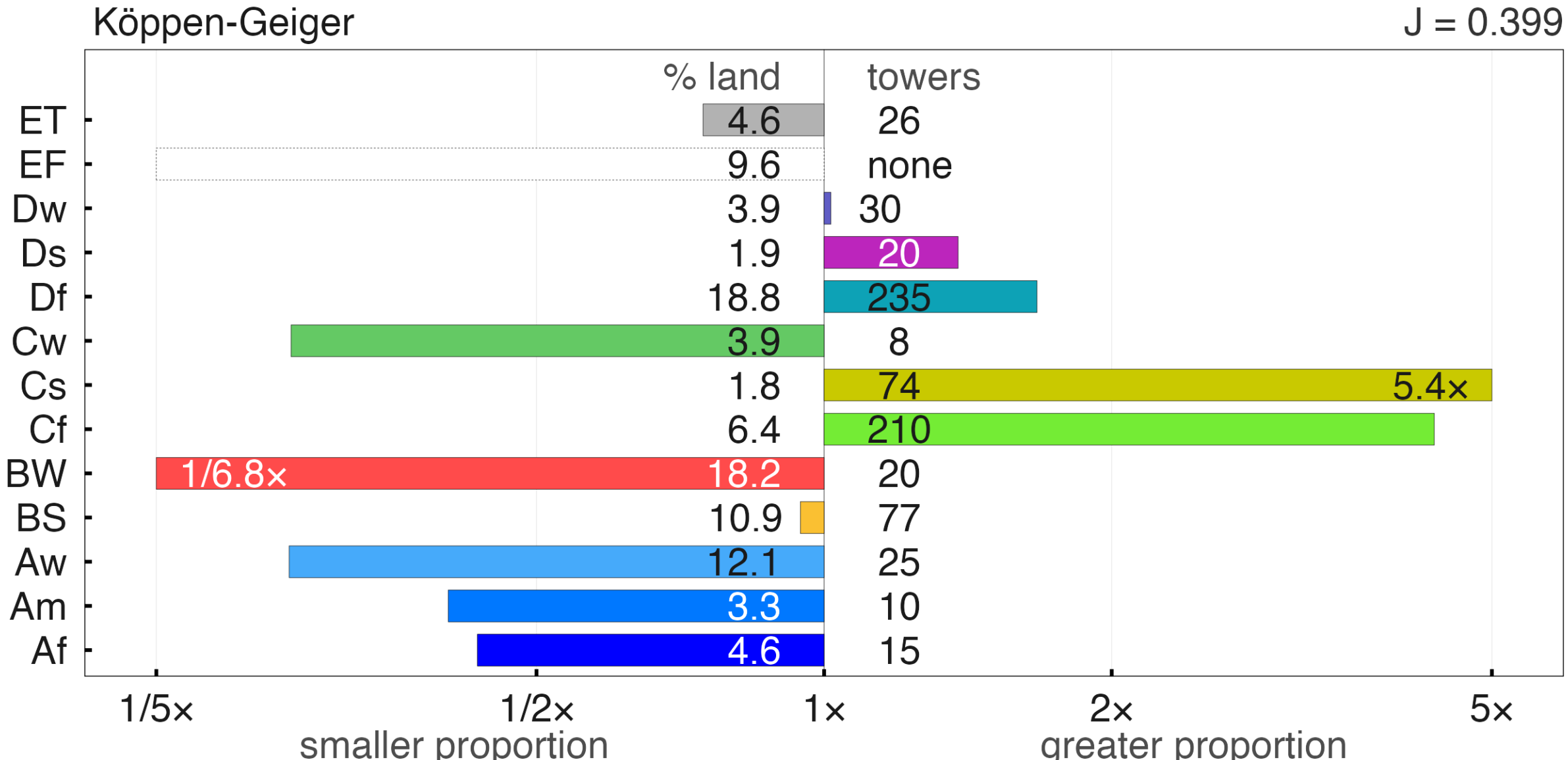

FLUXNET Shuttle median ET
± SD (mm yr−1)
1 000
750
500
250
0
0
250
500
750
1 000
FLUXNET2015 median ET
± SD (mm yr−1)
EBF
WSA
GRA
CRO
DBF
MF
WET
ENF
SAV
OSH

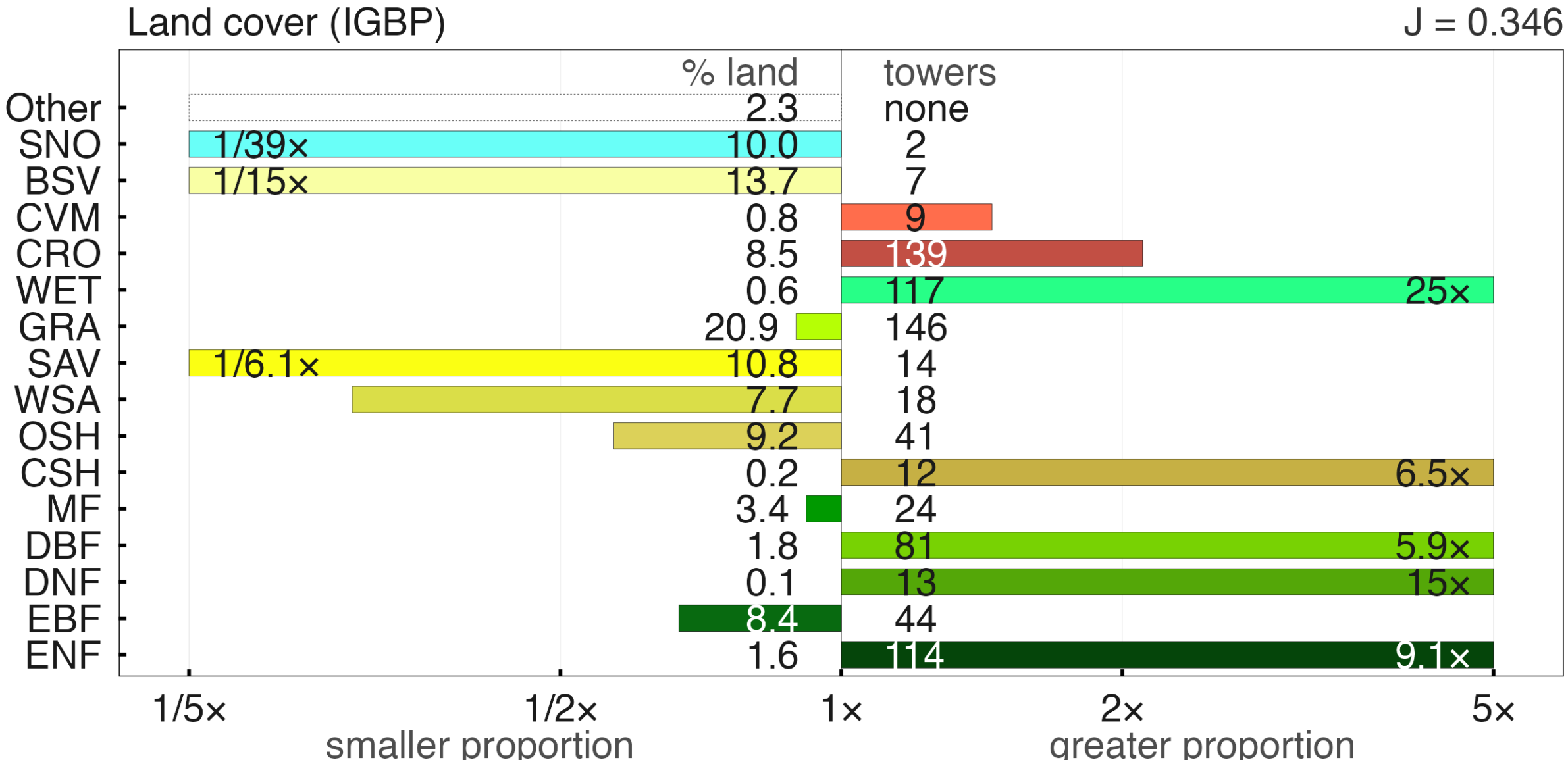

FLUXNET Shuttle median H
± SD (W m−2)
FLUXNET2015 median H
± SD (W m−2)
0
20
40
60
80
WSA
SAV
OSH
EBF
ENF
MF
CRO
DBF
GRA
WET

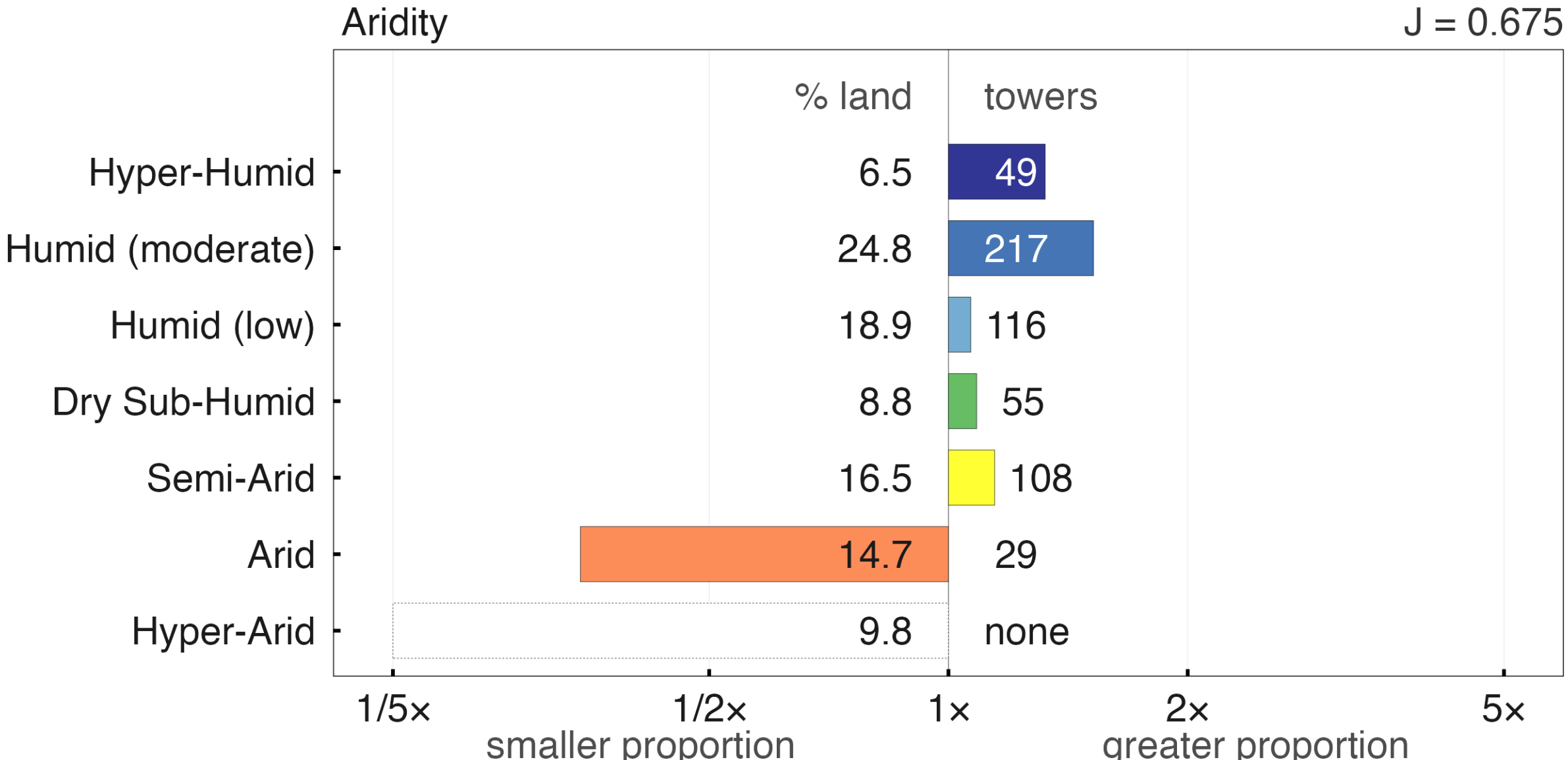

NEE (VUT)
NEE (CUT)
LE
H
Cumulative share
1.0
0.5
0.0
0.50
0.75
1.00
Annual QC flag (fraction measured or good-quality gap-filled)

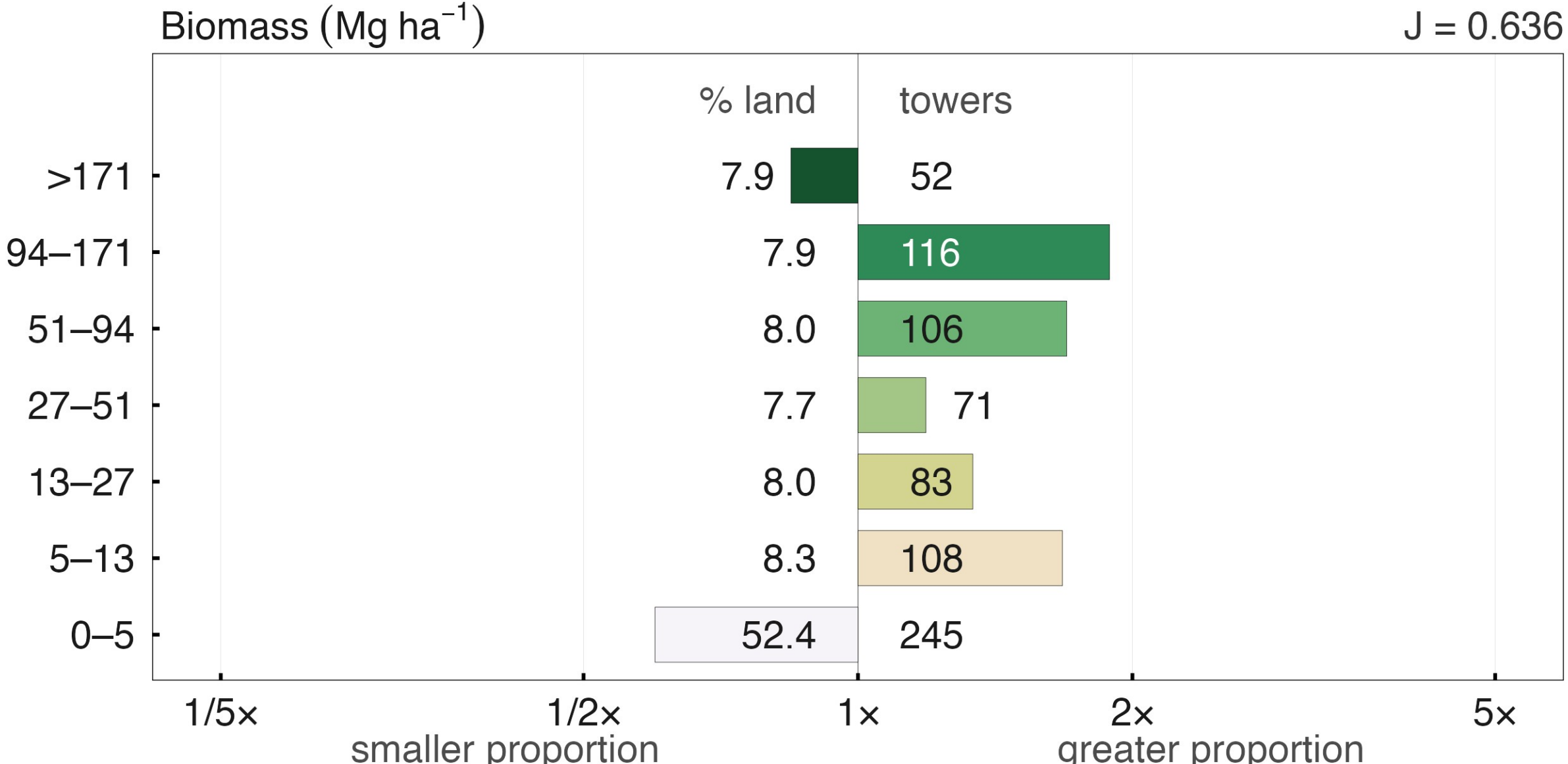

VUT
CUT
100
10
1
Uncertainty (g C m^{-2} yr^{-1}, log$_{10}$)
random
u*-threshold
joint
Uncertainty term

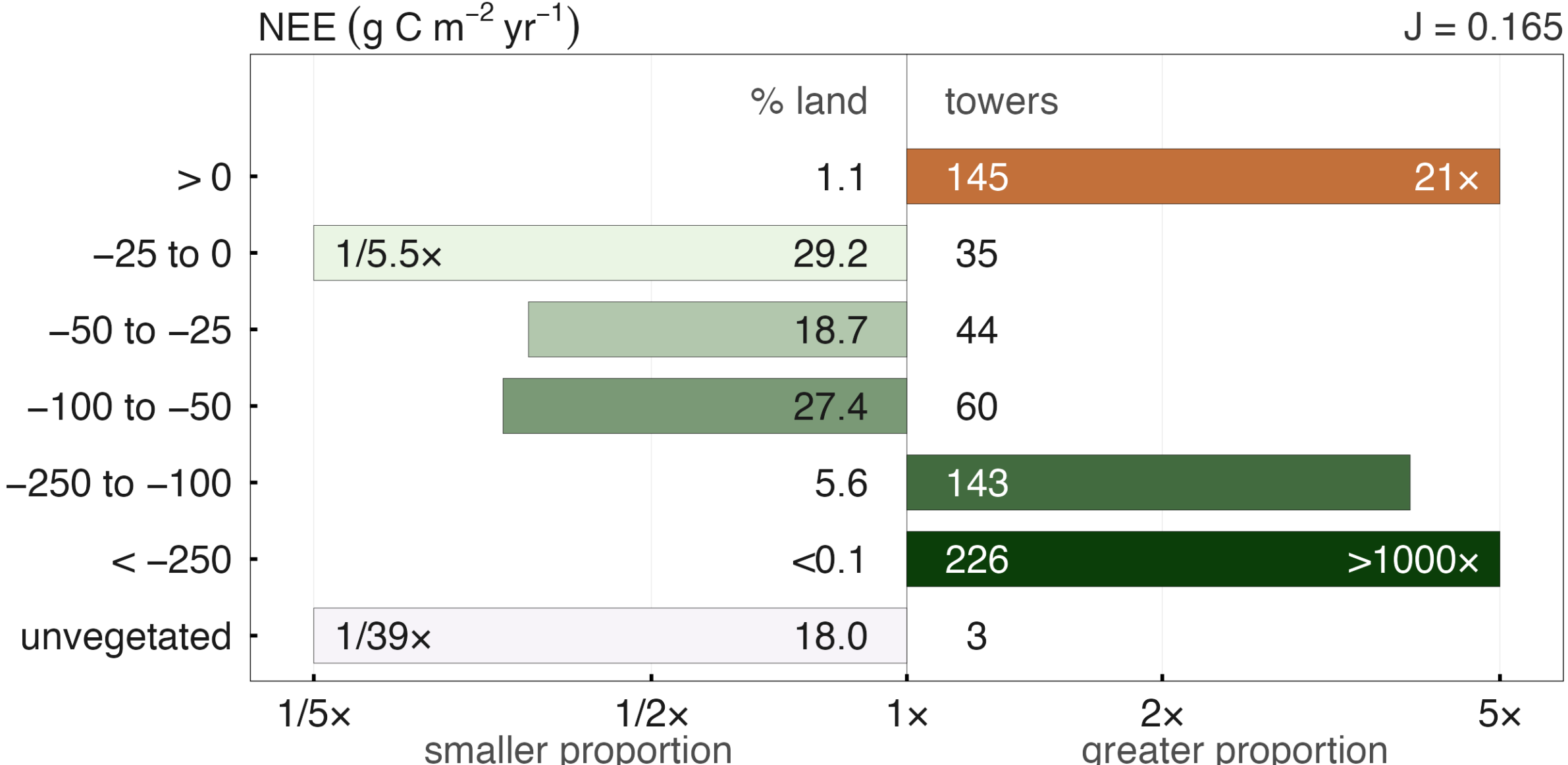

Site-years
1000
500
0
−100
0
100
NEE$_{VUT}$ - NEE$_{CUT}$ (g C m^{-2} yr^{-1})

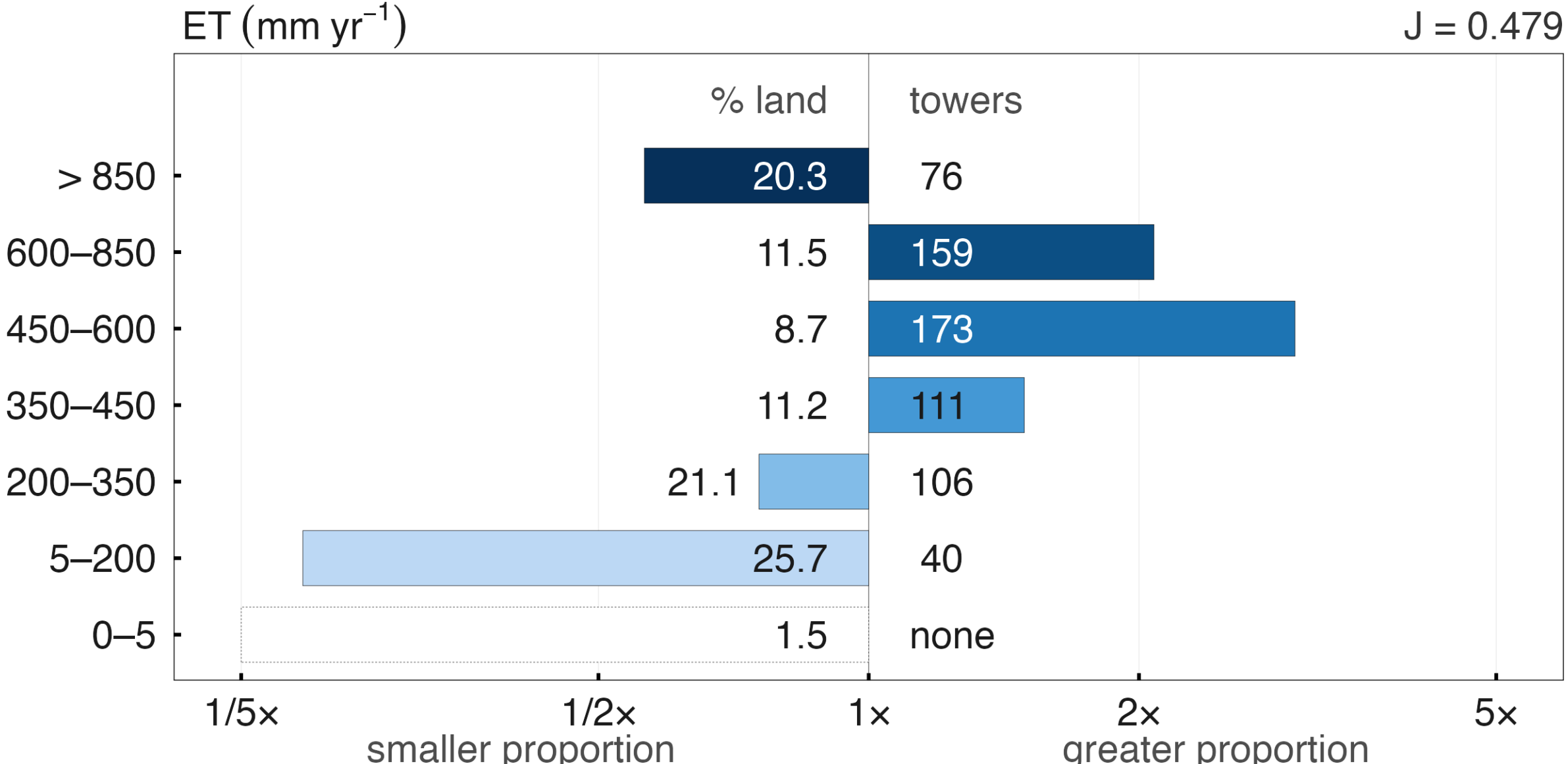

Köppen-Geiger
J = 0.399
% land
towers
ET
EF
Dw
Ds
Df
Cw
Cs
Cf
BW
BS
Aw
Am
Af
4.6
9.6
3.9
1.9
18.8
3.9
1.8
6.4
18.2
10.9
12.1
3.3
4.6
26
none
30
20
235
8
74
210
20
77
25
10
15
1/6.8×
5.4×
1/5×
1/2×
1×
2×
5×
smaller proportion
greater proportion


Land cover (IGBP)
J = 0.346
% land
towers
Other
2.3
none
SNO
1/39×
10.0
2
BSV
1/15×
13.7
7
CVM
0.8
9
CRO
8.5
139
WET
0.6
117
25×
GRA
20.9
146
SAV
1/6.1×
10.8
14
WSA
7.7
18
OSH
9.2
41
CSH
0.2
12
6.5×
MF
3.4
24
DBF
1.8
81
5.9×
DNF
0.1
13
15×
EBF
8.4
44
ENF
1.6
114
9.1×
1/5×
1/2×
1×
2×
5×
smaller proportion
greater proportion


Aridity
J = 0.675
% land
towers
Hyper-Humid
6.5
49
Humid (moderate)
24.8
217
Humid (low)
18.9
116
Dry Sub-Humid
8.8
55
Semi-Arid
16.5
108
Arid
14.7
29
Hyper-Arid
9.8
none
1/5×
1/2×
1×
2×
5×
smaller proportion
greater proportion


Biomass (Mg ha^{-1})
J = 0.636
% land
towers
>171
94–171
51–94
27–51
13–27
5–13
0–5
7.9
52
7.9
116
8.0
106
7.7
71
8.0
83
8.3
108
52.4
245
1/5×
1/2×
1×
2×
5×
smaller proportion
greater proportion


NEE (g C m^{-2} yr^{-1})
J = 0.165
% land
towers
> 0
1.1
145
21×
–25 to 0
1/5.5×
29.2
35
–50 to –25
18.7
44
–100 to –50
27.4
60
–250 to –100
5.6
143
< –250
<0.1
226
>1000×
unvegetated
1/39×
18.0
3
1/5×
1/2×
1×
2×
5×
smaller proportion
greater proportion


ET (mm yr^{-1})
J = 0.479
% land
towers
> 850
20.3
76
600–850
11.5
159
450–600
8.7
173
350–450
11.2
111
200–350
21.1
106
5–200
25.7
40
0–5
1.5
none
1/5×
1/2×
1×
2×
5×
smaller proportion
greater proportion